

Alejandro Martín Auden

QUANTUM COMPUTING | QUANTUM MACHINE LEARNING | SOFTWARE ENGINEERING

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Profile

M.Sc. student in Quantum Computing with a B.S. in Computer Engineering and two years of professional software engineering experience. I work on both sides of the problem: hybrid quantum-classical models and quantum algorithms on one hand, production backend and data systems on the other. My research so far has centred on quantum algorithms and on the evaluation of entangled, non-classical states, developed through my bachelor's and master's theses, complemented by summer-school training on real quantum hardware and HPC platforms. Looking to apply quantum computing to real industrial problems in an applied R&D environment.

Education

Universidad Internacional de la Rioja, M.S. in Quantum Computing Mar 2025 – Present

- **Coursework:** Quantum Mechanics, Information Theory, Quantum Algorithms

Universidad de Sevilla, B.S. in Computer Engineering Sept 2019 – Jun 2024

- **Coursework:** Computer Architecture, Software Architecture, Distributed Systems

Experience

Quantum Computing Intern, Quantum Realm Games – Remote May 2026 – Present

- Developing a quantum machine learning framework and implementing quantum algorithms in Python.
- Designing and implementing evaluation metrics for quantum game states, working with superposition and entanglement.
- Analysing quantum game datasets to identify patterns and guide heuristics, and communicating the findings.

Associate Analyst Developer, FNZ – Brno, Czechia Jul 2024 – Nov 2025

- Built and optimised backend systems and data pipelines in C# and .NET for large-scale financial infrastructure, cutting report generation time by up to 50%.
- Worked in a distributed Agile team using Git, with code review, testing and maintainable architecture as everyday practice.

Database and IT Assistant, Ayesa – Sevilla, Spain Sep 2022 – Jan 2023

- Designed database mechanisms for accessing and visualising large datasets, and automated repetitive data operations.

Research

Quantum Neural Network for Quantum Chess

2026 – Present

Master's Thesis, UNIR

Python

- A hybrid reinforcement-learning architecture that optimises variational quantum circuits through self-play in order to navigate entangled board positions.
- Involves working directly with superposition and entanglement in multi-qubit states, and with the measurement statistics used to evaluate them.

Quantum Chess: implementation and heuristics

2023 – 2024

Bachelor's Thesis, Universidad de Sevilla

Python

- Implemented the quantum game designed by Christopher Cantwell (2016) and introduced new heuristics for evaluating its non-classical states.
- Proposed a method for evaluating quantum game states built on classical chess engines, addressing how to score positions held in superposition.

Other projects: Trakeo  — full-stack finance platform (Next.js, TypeScript, PostgreSQL/Prisma, event-driven background processing).

Research Training

Quantum Applications across Digital and Analog Architectures

JUNIQ-EPIQ Summer School, Forschungszentrum Jülich

August 2026

Non-classicality in Quantum Theory and Beyond

Jyväskylä Summer School

August 2026

Quantum Cellular Automata

Jyväskylä Summer School

August 2026

Machine Learning Specialization

DeepLearning.AI and Stanford University

Sept – Nov 2023

Skills

Quantum computing: Quantum algorithms, hybrid quantum-classical models, variational circuits; Qiskit, PennyLane, Quirk; validation on simulators and real quantum hardware.

Foundational knowledge: quantum information theory and entropies; non-classicality and Bell inequalities; quantum cryptography and quantum communications.

Programming: Python, C#, .NET, Java, C++, SQL; Git, CI/CD, REST; Agile delivery, code review and testing.

Languages: Spanish (native), English (fluent), Czech (beginner), French (basic).